

NEHAL ABDUL REHMAN

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RESEARCH INTERESTS

- On-device AI perception and deployment on resource-constrained edge hardware
- Embedded systems and real-time firmware for IoT and sensing platforms
- Real-time embedded control and sensor interfacing for robotic and industrial systems
- Multisensor integration for autonomous platforms (vision and IMU)

EDUCATION

B.E. Industrial Electronics Engineering

[Dec/2018 – Jan/2023]

Institute of Industrial Electronics Engineering (IIEE), affiliated with NED University of Engineering & Technology (NEDUET), CGPA 3.58/4.00

- **Ranked 3rd** of 50 in the program. **Washington Accord**-accredited (PEC), OBE program; **Medium of Instruction (MOI):** English.
- **Capstone/Final Year Project (FYP):** Real-time 2D surface detection and inspection (Python/OpenCV, Raspberry Pi 4). Supervisor: Engr. Dr. Farah Haroon (Principal, IIEE).
- **Coursework:** Digital Signal Processing, Embedded Systems, Microprocessor Architecture & Programming, Industrial Process Control, Electronic Motor Drives, Signals & Systems, Digital Logic Design, Linear Algebra, Numerical Methods.

PUBLICATION

Rehman, N.A.; Haroon, F. Adaptive Gaussian and Double Thresholding for Contour Detection and Character Recognition of Two-Dimensional Area Using Computer Vision. Engineering Proceedings 2023, 32(1), 23 (MDPI). Presented at the **2nd International Conference on Emerging Trends in Electronic and Telecommunication Engineering**, Karachi, 15–16 March 2023. doi:10.3390/engproc2023032023

- Compared Adaptive Gaussian Thresholding with Canny edge detection (high/low thresholds) for contour measurement and character recognition on high- and low-quality images; Adaptive Gaussian Thresholding outperformed Canny on both sharp and blurry images.
- First and corresponding author. Led problem formulation, experimental design, and implementation, extending the work from my FYP/capstone; wrote the manuscript and presented it at the conference. Supervised by Engr. Dr. Farah Haroon.

SELECTED RESEARCH & TECHNICAL PROJECTS

Real-Time Driver Drowsiness Detection (YOLOv5s)

NCAI | Smart City Lab

- Built a custom labeled dataset, trained a YOLOv5s detector, and deployed it on an NVIDIA Jetson AGX Xavier for real-time evaluation. (mAP > 0.9, 30 FPS)

Real-Time 2D Surface Detection & Inspection (Capstone/FYP)

Python, OpenCV, Raspberry Pi 4

- Implemented perimeter detection, feature matching and Tesseract text recognition, validated on a purpose-built conveyor rig. (97% accuracy, 15 FPS on Raspberry Pi 4)

Multi-Camera Driver for NVIDIA Jetson Orin

Embedded Linux (Linux for Tegra), C/C++

- Extended a single-camera stack to multiple MIPI-CSI lanes and resolutions for the AP1302 ISP; modified device-tree bindings for simultaneous capture; automated configuration with Python. (4 cameras, 720p to 1080p, 30 fps)

Low-Latency Wireless MIPI-CSI Video (RK3566)

Rockchip/ Buildroot / Firefly SDK

- GStreamer pipeline with hardware H.264 encode/decode, LT7911D MIPI-CSI-to-HDMI bridge and 5 GHz AP6275S link with custom driver bring-up. (HD stream transmitter and receiver)

Weed Bot 2.0: Autonomous Agricultural Robot

NCAI | Smart City Lab

- Controlled BLDC drive motors via Arduino and 350 W EV controllers; validated ATmega-based control subsystems.

Also: Allwinner V3s multi-channel smart meter (Qt GUI, MQTTs to AWS IoT Core); STMicroelectronics-STM32F0 non-blocking sequential controller; TI C2000 AC motor controller with OCP/OVP fault recovery; PLC–VFD Modbus control; ESP32 BLE smart charger (Wi-Fi provisioning, EEPROM data logging, JSON over WebSocket); Arduino PID temperature controller; Chebyshev Type II IIR speech-band filter (MATLAB).

PROFESSIONAL EXPERIENCE

Embedded IoT Engineer, Pouлта, Inc.

Jan 2026 – Present

- RTOS firmware (C/C++, ESP-IDF/ESP32) for IoT edge nodes; OTA update pipelines; CI with unit testing and linting through CD with hardware-in-the-loop testing (GitHub Actions).

Embedded Firmware Engineer, eSpark Consulting Group

2024 – 2026

- Multi-platform systems (STM32, TI C2000, ESP32, RK3566, Allwinner V3s, NXP i.MX93) for smart metering and motor control; Qt/LVGL interfaces; secure device-to-cloud data paths (MQTTS to AWS IoT Core, SFTP); debugged memory leaks and corruption with J-Link and XDS100.

Junior Embedded Firmware Developer, UTF-Labs

2022 – 2024

- Optimized RTOS firmware for memory and race conditions; IMU, PMIC and display drivers; BLE 5.0 dual-role (central and peripheral) with 2M PHY for high-speed transfer of large log datasets.

Robotics & Embedded Intern, NCAI | Smart City Lab

Jan – Mar 2023

- Edge-AI perception and field-robot control (see projects above).

IIoT & Industry 4.0 Intern, Mindstorm Engineering

Sep – Oct 2021

- PLC programming (Ladder, FBD, STL) and SCADA/HMI on Siemens S7-300/S7-1200 (TIA Portal).

HONORS & SCHOLARSHIPS

- Semester merit scholarship (four times); position holder in the 3rd, 5th, 6th and 7th semesters; 3rd position in the degree program.
- 3rd highest score nationwide in the NTS-IIIE entry test (2018).
- Amal Academy Career-Prep Fellowship (Stanford-funded), selected competitively from 4,500+ applicants (2021).

TECHNICAL SKILLS

Modeling & simulation: MATLAB, Simulink, LabVIEW, Proteus, Multisim, Gazebo, JMAVSim

Vision & AI: OpenCV, YOLO, Tesseract, GStreamer, NVIDIA Jetson (AGX Xavier, Orin), Raspberry Pi 3/4

Embedded & languages: C++, Embedded C, Python; FreeRTOS, ESP-IDF, STM32Cube, Embedded Linux (Yocto, Buildroot); J-Link, XDS100; Qt, LVGL

Control & protocols: PID control, PLC/SCADA/HMI, Variable Frequency Drive (VFD), BLE, MQTT(s), HTTP(s), I2C, UART, SPI, ZWave, WiFi, Bluetooth

LANGUAGES

English: professional proficiency - language of instruction throughout education [MOI certificate available on request] | Chinese: HSK I & II

REFEREES

Engr. Dr. Farah Haroon, Principal, IIEE. Final-year project supervisor, instructor for communication systems and digital signal processing courses, and paper co-author. farah@iiee.edu.pk

Engr. Dr. Sajid Hussain, Assistant Professor, IIEE. Instructor for microprocessor architecture & programming and object oriented programming courses. sajid@iiee.edu.pk

Dr. Engr. Asif Ahmed Memon, Assistant Professor, IIEE. Instructor for embedded systems, digital logic design and electrical machines courses. asifmemon@iiee.edu.pk